Prototype refinement

Iterative refinement

Support prototypes

Soft assignments

Weighted soft prototypes

Refined prototypes

Blend

Weighted mean

Unlabeled features

Retain support anchors

Sharpening

Fixed signed gates

Class prompts P

Node-wise correction

Post-training: calibrated readout

Attribute similarity

Neighbor selection

Attribute graph

a

b

c

Centered two-hop features

Teacher features from M1

Exclude self

Similarity weights

Shared frozen graph encoder

Input graph

Layer 1

LeakyReLU

Hidden-feature correction

Layer 2

Adapted representations

Prototype classifier

Initial probabilities

From Prompt scores s

Fixed W

Local agreement → node temperature

Initial classes

Weighted by W

$a_i = \sum_j W_{ij} \mathbf{1}[c_j = c_i]$

$T_i = 1 + 3(1 - a_i)$

$F_{0,i} = \mathrm{softmax}(s_i / T_i)$

Low agreement → soften

Prompt learning: direction and norm adaptation

Support loss

Shared gradient G

Rescaled direction $\tilde{D}$

Same ray; only the length changes

Shampoo direction D

Common origin

Adam reference A

Use its norm, not its direction

$\|\tilde{D}\|_F = \|A\|_F$

Match the whole-matrix norm

$P \leftarrow P - \eta \tilde{D}$

Parameters move in the negative direction

Dashed-circle radius = Adam norm

Current probabilities

Neighbor propagation

Updated probabilities

20 iterations

Calibrated F_0

10%

Weighted neighbor probabilities

90%

Updated class probabilities

Clamp support labels

Predicted classes

Frozen backbone

Support node

Only P is updated

CGRPrompt: Class-Guided Residual Prompt

One class-residual prompt: construction, learning, readout

Temperatures are computed once; backbone and teacher features stay fixed